LAB REPORT

Group-	IPE-06
Group Members-	Tahamid Arshil (45) Jabir Ahmed (35) Tarif Shahriar (16) Syeda Tahiya Hassan (25) Moho Hassan (06)
Course Code-	ME 4408
Name of the Project-	Temperature, humidity and smoke sensing with temperature and smoke control using light, fan and, servo motor.
Date of Submission-	22/03/2022

Objectives:

- To detect temperature and smoke through the temperature-humidity (HSM-20G) sensor.
- To use a data collection system to regulate and save the temperature and humidity data that has been measured.
- Using a sensor, detect the presence of smoke in a place and remove it with the opening of a lid.

The overall purpose of this project is to learn how to detect different factors using sensors, develop a logic using software to control the parameters using desirable components, and ultimately equip actuators to execute desired jobs.

Theory:

Mechatronics is an integrated area of engineering that combines mechanical and electrical engineering and computer science. A typical mechatronic system hoists signal from the environment, processes them to produce output signals, converting them for example into forces, motions and actions.

It is the augmentation and execution of mechanical systems with sensors and microcomputers which is the most crucial aspect. The fact that such a system lifts changes in its environment by sensors, and responds to their signals using the proper information processing, makes it non-identical from conventional machines.

Examples of mechatronic systems are robots, digitally controlled combustion engines, machine tools with self-adaptive tools, contact-free magnetic bearings, automatic guided vehicles, etc.

In our instrumentation project, we are using temperature and humidity sensor. The module of HSM-20G is essential for those applications where the relative humidity can be converted to standard voltage output.

Its applications are as follows:

- Humidifiers and dehumidifiers
- Air-conditioner
- Humidity data loggers
- Automotive climate control
- Other applications

Equipment Used:

The following components were used to complete the project:

1. Arduino Mega with 12V adapter

2. Temperature-Humidity Sensor (HSM-20G)

3. Smoke sensor

4. 2 channel 5V relay

5. CPU cooling fan

6. Light Holder, 60 W light, Electric plug and 5 feet cable

7. Servo Motor

8. 2 pieces of resistor (10KΩ and 100KΩ)

9. Capacitor 47 μF

10. Potentiometer (10K)

11. LED

12. Breadboard (1 small)

13.Jumper wire set (MM, MF, FF)

14. Cotton buds.

15. 5 Transparent Acrylic Boards, Hacksaw blade.

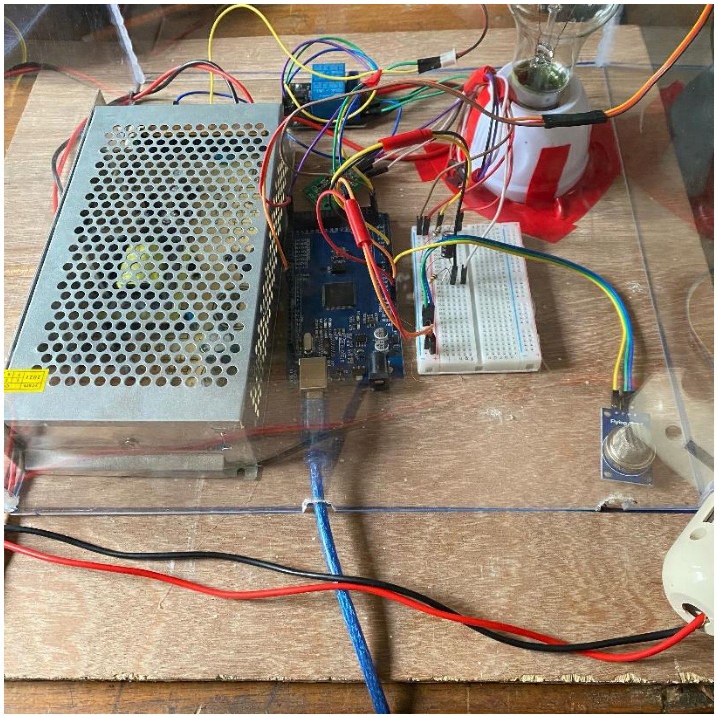
16. Glue Gun

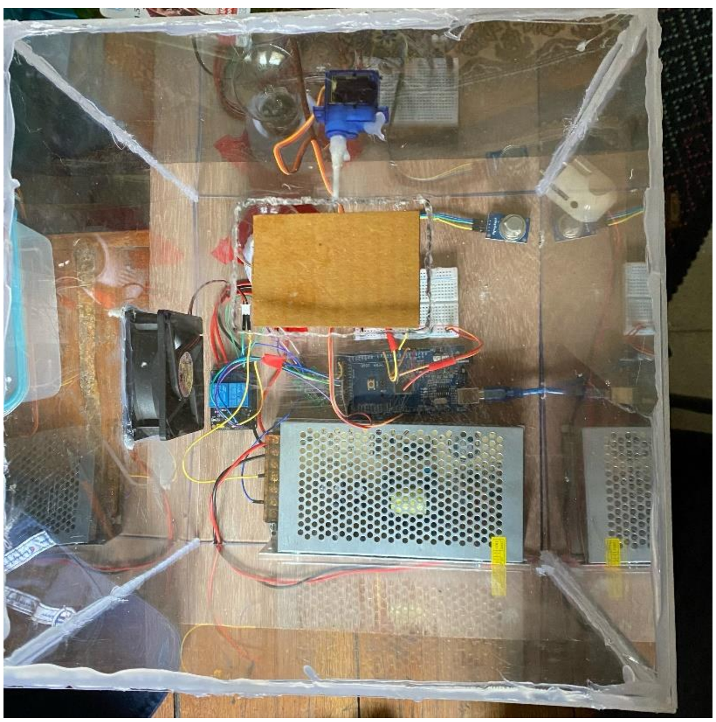
17. Plywood board for base

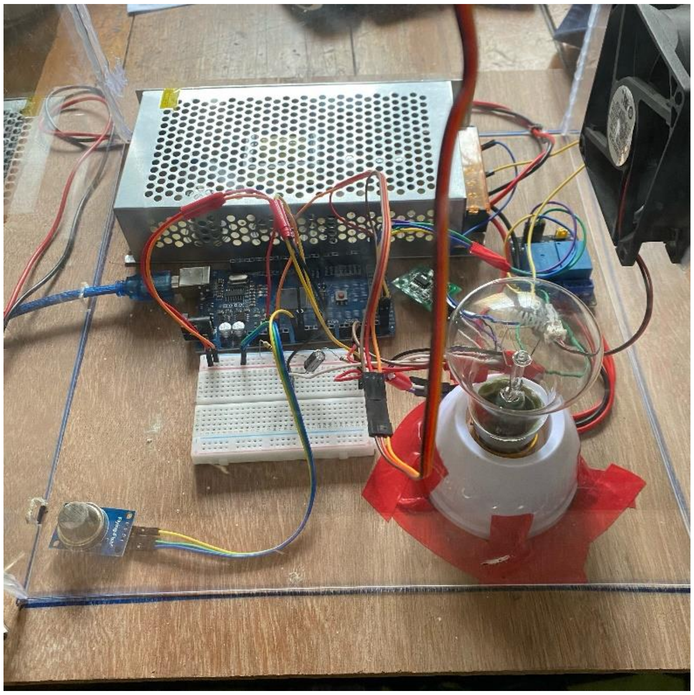
18. White Scotch-tape, board pin.

Setup Design:

The images below can help us visualize the project setup from the front, top and side.







Some design intricacies have been explained below.

Servo Connection:

The servo motor was coupled with a cotton bud and a piece of thin cardboard so that the small amount of torque created could support the door and function as a smoke route.

Box Compartment:

The box was made with a control volume so that the temperature could be varied and regulated. The box has two openings: one for the smoke lid and one for the exhaust of air for cooling. The smoke cover was made out of cardboard to keep it as light as possible while maintaining stiffness. It's also controlled by a tiny servo motor and a cotton bud, because of its lightweight properties.

The relay is used to isolate the low voltage Arduino and electronics circuit from the high voltage mains supply.

The fan has been configured as an intake fan. Because the purpose of employing the fan is to radiate heat using the air from the fan in the control volume, this is done. Using an intake fan, the heated air is radiated out of the control volume, cooling the box. Cool air enters the box through the intake, transferring heat to the air and keeping the box cool.

Because of its stiffness, plywood was employed as a basis for our experiment. Initially, we considered utilizing acrylic, but it proved to be excessively flexible and susceptible to breaking. As a result, plywood proved to be the greatest alternative since it was inexpensive, solid, and simple to work with.

Circuit:

In our project we have used Temperature – Humidity (HSM-20G) with 4 pin crow which has 4 output points and they are for the temperature output, humidity output, voltage output (5V) and the ground output.

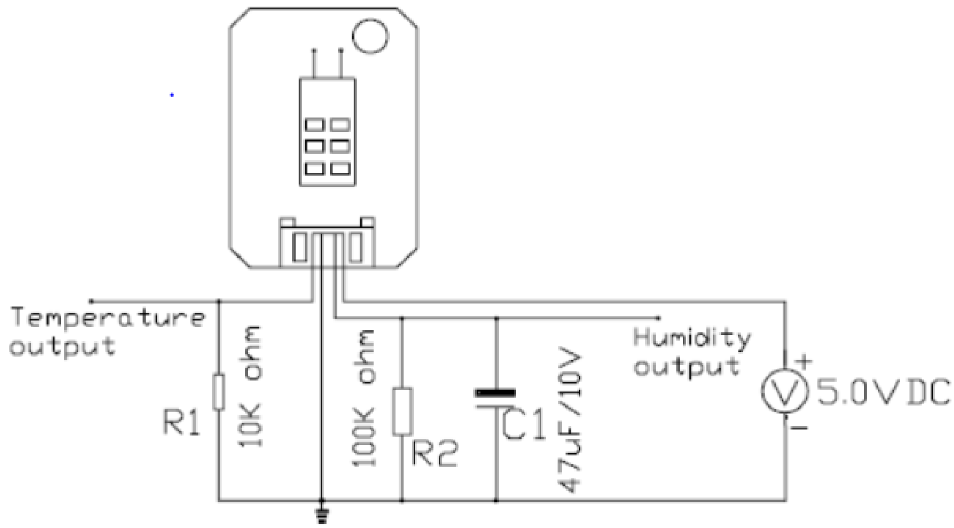


Figure 1: Circuit Diagram

From the circuit diagram we can see that the temperature output part from the sensor is connected to 10kΩ resistor in series connection and the humidity output is connected with 100kΩ resistor and 47µF capacitor in parallel connection. Here we have applied the voltage divider rule. The outcome of dividing the input voltage among the divider's components called voltage division. The resistance R of an object also varies with temperature:

$R=R_0(1+\alpha\Delta T)$

Where,

R_0 is the original resistance and R is the resistance after the temperature change.

For the humidity,

STANDARD CHARACTERISTICS

%RH	10	20	30	40	50	60	70	80	90
OutpotV	0.74	0.95	1.31	1.68	2.02	2.37	2.69	2.99	3.19

The positive output from the sensor is connected with the +5V channel in the Arduino Mega. In the breadboard we have connected the circuit according to the diagram where the ground output is grounded.

In the Arduino Mega 2560 there are three channels – analog, digital and PWM (pulse with modulation). The PWM is the regulate type control here. The voltage output from the sensor is connected to the +5V of the Arduino in the digital channel through breadboard and the temperature and humidity output is connected to the A_0 and A_1 channel. The negative voltage output is grounded again.

From the relay we can see that there are two channels with three points each and they are – NO (Normally open), NC (Normally closed) and COM (Common). The positive terminal of the light is connected to the NO port of one channel and the negative terminal is connected to the N port of the power supply. And here in the N port the negative terminal from the holder is also attached. On the other hand, the positive terminal from the power supply which is known as L port is connected to the COM port of the relay and there in the L port the positive point from the holder is also linked.

For the fan the positive part is connected in the 2nd relay channel with the NO port and the negative port from the fan is connected to the -V port from the power supply. The +V port from the power supply is linked with the COM channel again.

Here the relay is used to regulate the voltage and it sends the voltage to the Arduino. The electromagnetic induction principle governs the operation of a relay. When a current is applied to an electromagnet, it creates a magnetic field around it. Relays can eliminate the need for expensive and space-consuming high-amperage wire and switches. Switching to relays in your electronic systems can, for example, reduce the size or weight of a casing or allow manufacturers to fit more functionality into the same amount of space.

On the other side of the relay, we can see 4 other ports which are: GND, IN1, IN2 and VCC. the 4 ports are respectively connected to the GND, Temperature (A_0), Humidity (A_1) and the +5V from the Arduino mega. Here from the Arduino, we have to generate power to breadboard. Arduino sends the commands to relay.

From the servo motor, we can see there are three wires – Orange one for the signal, black one for the ground connection and the red one for the positive connection. The signal wire is attached to the PWM channel from the Arduino and the GND and Positive port is connected to the GND and POSITIVE port respectively from the digital channel of the Arduino mega.

For the smoke sensor there are also 4 points which are A_0, D_0, GND & VCC. The A_0 one is connected to the analogue channel of the Arduino and the GND and the VCC are connected to GND and +5V respectively from the breadboard.

Here we have used the PID controller and Proportional, Integral, and Derivative is the acronym for Proportional, Integral, and Derivative. PID control uses a control loop feedback mechanism to give a continuous variation of output to properly regulate the process, eliminating oscillation and enhancing process efficiency.

Software:

Codes used:

For temperature and humidity-

R=(5.0-V)*10.0/V

Temp=281.583*(1.0230^(1.0/R))*(R^-0.1227)-150.6614

Hum=(31*H)-12

For servo angle and smoke-

in_min=VS_min

;in_max=VS_max

out_min1=1350

out_max1=2500

out1=out_min1+((out_max1-out_min1)/(in_max-in_min)*(in-in_min))

servo_us=out1

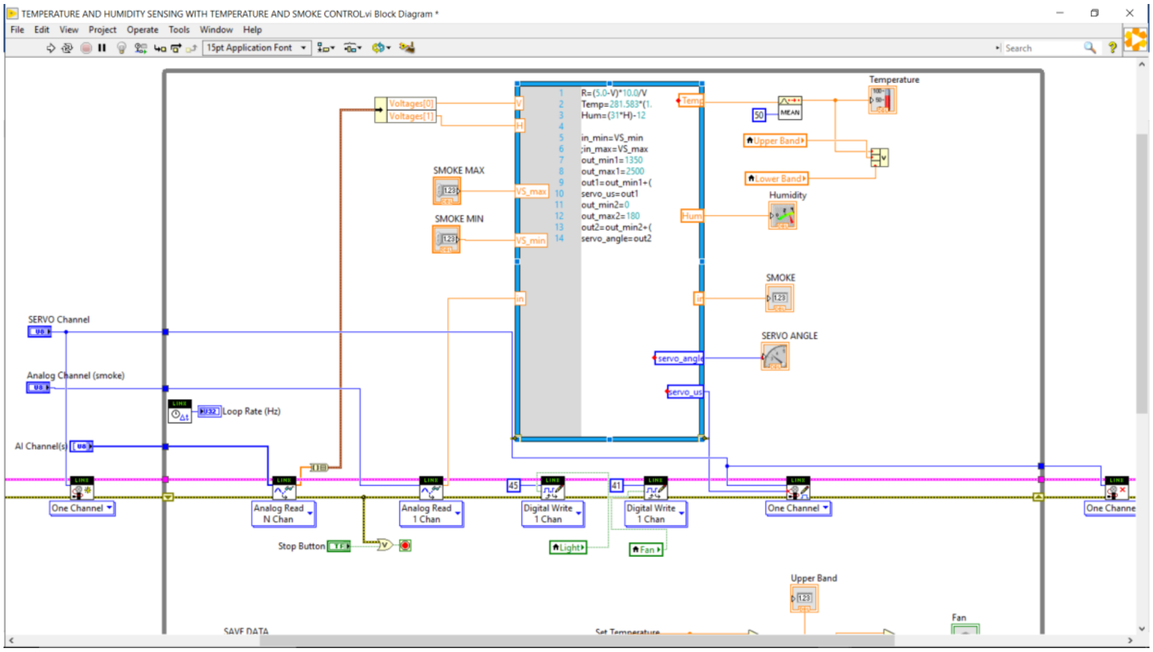
out_min2=0

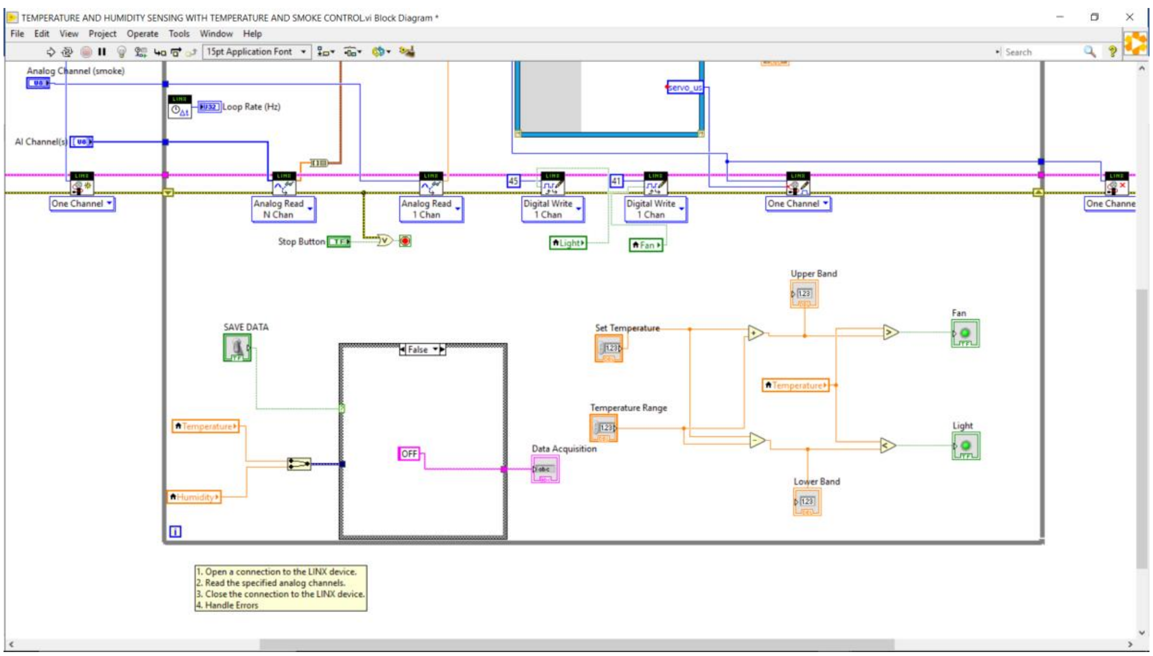
out_max2=180

out2=out_min2+((out_max2-out_min2)/(in_max-in_min)*(in-in_min))

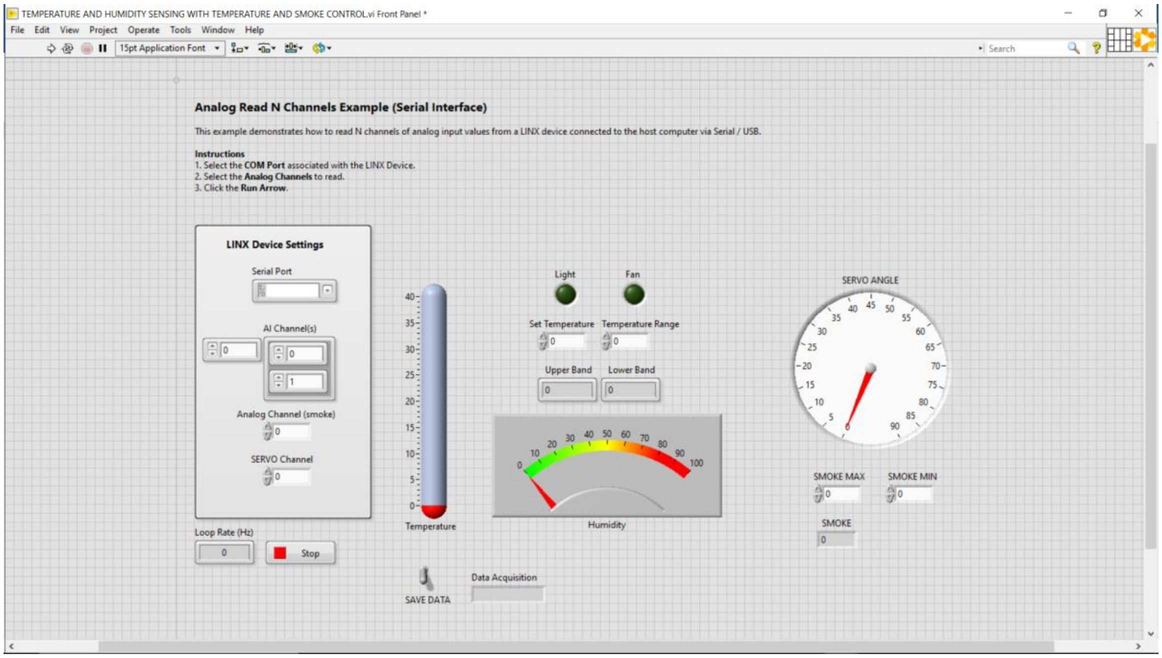
servo_angle=out2

Software Block Diagram:





Software front panel:



Discussions:

Difficulties faced:

- The sensors utilized in this project were quite sensitive, and the primary issue we encountered was that our temperature-humidity sensor failed to function correctly at one point, necessitating the replacement of the sensor.
- The circuit that we were following was not accurately supplied input, resulting in an incorrect temperature measurement.
- We also had a difficulty with our relay, which necessitated changing it to a 5V one, and the servo motor was not accepting readings or functioning properly.
- We couldn't cut the acrylic board neatly since it was thick enough. After that, we chose to utilize the soldering iron. We used an anti-cutter to cut through the acrylic board after heating the soldering iron.

Cost-Estimation:

Sl. no.	Component	Quantity	Price (BDT)
1	Arduino Mega with 12V adapter	1	1090
2	Temperature – Humidity sensor (HSM -20G)	1	690
3	Smoke Sensor	1	120
4	2 Channel Relay	1	210
5	CPU Cooling Fan	1	70
6	Light Holder. 60W light, electric plug and 5 feet cable	1	135
7	Servo Motor	1	400
8	2 pieces of Resistor 10k and 100k. Capacitor 47µF	1	20
9	Potentiometer (10k) LED (1 or 2 color)	1	40
10	Breadboard - 1 small	1	65
11	Jumper Wire Set - MM MF and FF	5	450
12	Transparent Acrylic Boards, Hacksaw blade	5	1015
13	Glue Gun, glue stick	1	350
14	Plywood board for base	1	125
15	White Scotch-tape, board pin	1	30
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